EXPERIMENT 9: WIRELESS LAN CONFIGURATION

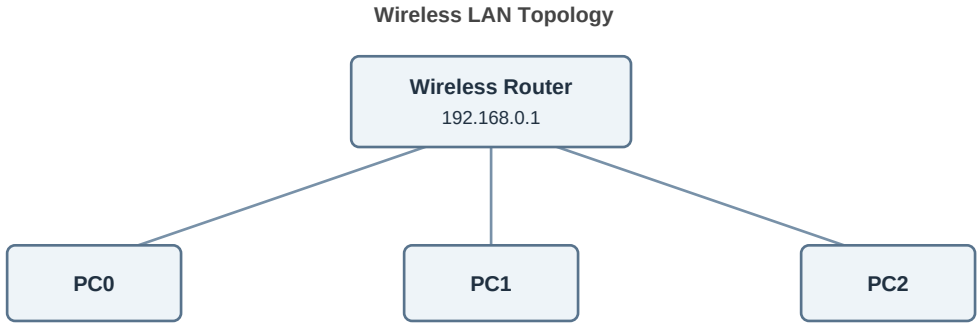
AIM

To configure a Wireless LAN in Cisco Packet Tracer by connecting three PCs to a Linksys wireless router, assigning static IP addresses, setting the SSID, and securing the wireless network.

REQUIREMENTS

Cisco Packet Tracer, one Linksys wireless router, and three PCs with wireless connectivity.

TOPOLOGY



Packet Tracer topology created for the experiment

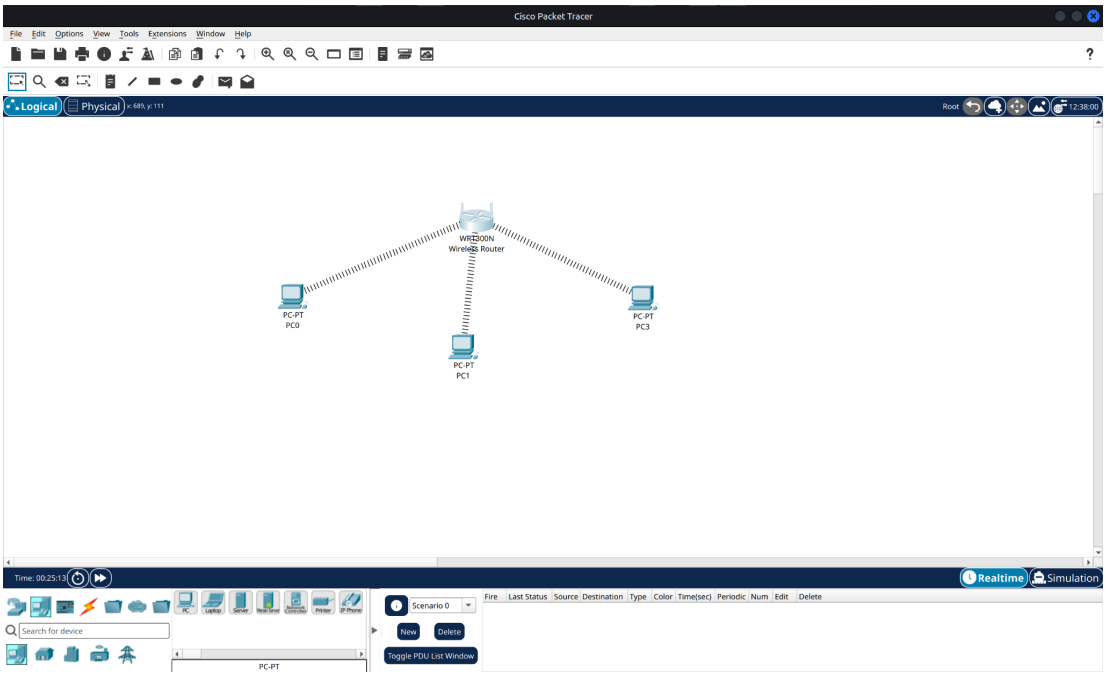


Figure: Wireless router with three PCs connected through wireless links in Cisco Packet Tracer.

CONFIGURATION

1. Open the wireless router and configure its administration username and password as **admin**, then save the settings.
2. Set the wireless SSID to **MotherNetwork**.
3. Under Wireless Security, select **WEP** and set Key 1 to **0123456789**. Save the settings.
4. Configure each PC with the static IP address, subnet mask, and default gateway shown below.
5. Connect each PC to the MotherNetwork wireless network using the configured key.

IP CONFIGURATION

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.0.1	255.255.255.0	—
PC0	192.168.0.2	255.255.255.0	192.168.0.1
PC1	192.168.0.3	255.255.255.0	192.168.0.1
PC2	192.168.0.4	255.255.255.0	192.168.0.1

WIRELESS SETTINGS

SSID: MotherNetwork **Security mode:** WEP **Key 1:** 0123456789

SAMPLE CONNECTIVITY TEST

From PC0, open Command Prompt and execute:

```
ping 192.168.0.3
```

Sample Output	Interpretation
Reply from 192.168.0.3: bytes=32 time<1ms TTL=128	PC1 is reachable from PC0.
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)	Wireless connectivity is successful.

OBSERVATIONS

The three PCs were configured with unique static IP addresses in the same subnet and connected to the wireless router using the MotherNetwork SSID and security key. Successful ping replies indicate communication between the wireless clients.

RESULT

A Wireless LAN was configured in Cisco Packet Tracer with three PCs connected to a Linksys wireless router. The SSID, wireless security settings, static IP addresses, and connectivity were configured and tested successfully.